

3.1.5 The use of finishes

Paper and board finishing

Content	Potential links to maths and science
Students should be aware of the ways that paper and board can be finished to enhance their appearance or for improved function. Specific finishes to include: • laminating • embossing • debossing • varnishing, UV varnishing and spot varnishing • foil blocking.	Ensure products are designed to take account of potential corrosion due to environmental factors.

Paper and board printing processes

Content	Potential links to maths and science
Students should be aware of the different types of printing processes and their suitability for specific products and scales of production. Specific processes to include: • screen printing • flexographic and offset lithographic printing • digital printing.	

Polymer finishing

Content	Potential links to maths and science
Students should be aware of the ways that polymers can be finished to enhance their aesthetics or for improved function. Students should be aware that some polymers are self-finishing and that this should be considered as a polymer finish. Specific finishes to include: • acrylic spray paints • thermoplastic elastomer.	

Content	Potential links to maths and science
Students should understand how pigments can be added to polymers in the moulding process, including: • gel coats when laminating GRP • smart pigments such as thermochromic or phosphorescent.	

Metal finishing

Content	Potential links to maths and science
Students should be aware of the ways that metals can be finished to enhance their appearance or prevent corrosion. Including applied finishes: • cellulose paint • acrylic paint • electro-plating • dip coating • powder coating • galvanising • sealants • preservatives • anodising • plating • coating • cathodic protection.	

Wood finishing

Content	Potential links to maths and science
Students should be aware of the ways that woods can be finished to enhance their appearance or prevent decay. Specific finishes to include: • applied finished: • polyurethane varnish • acrylic varnish • water based paints • stains • colour wash • wax finishes • danish oil • teak oil • pressure treating with chemical preservatives.	